

Distributed Systems

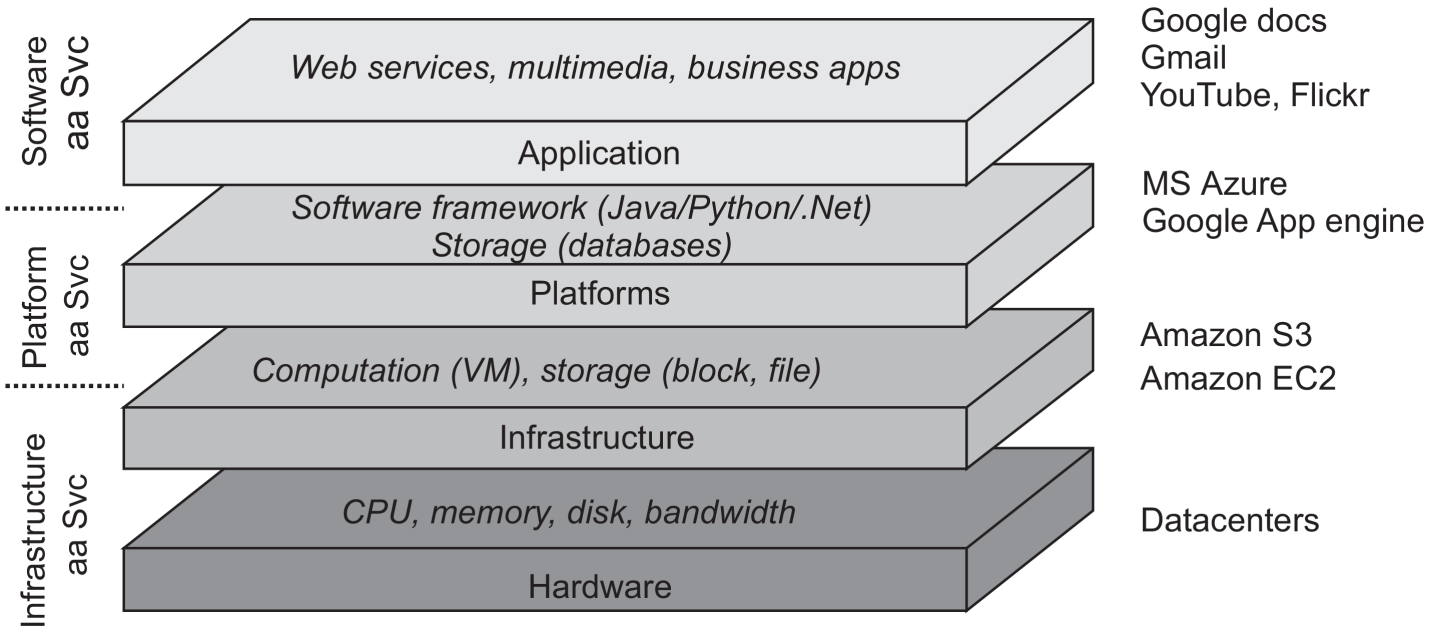
Lecture8: Hybrid System Architectures

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Cloud Computing

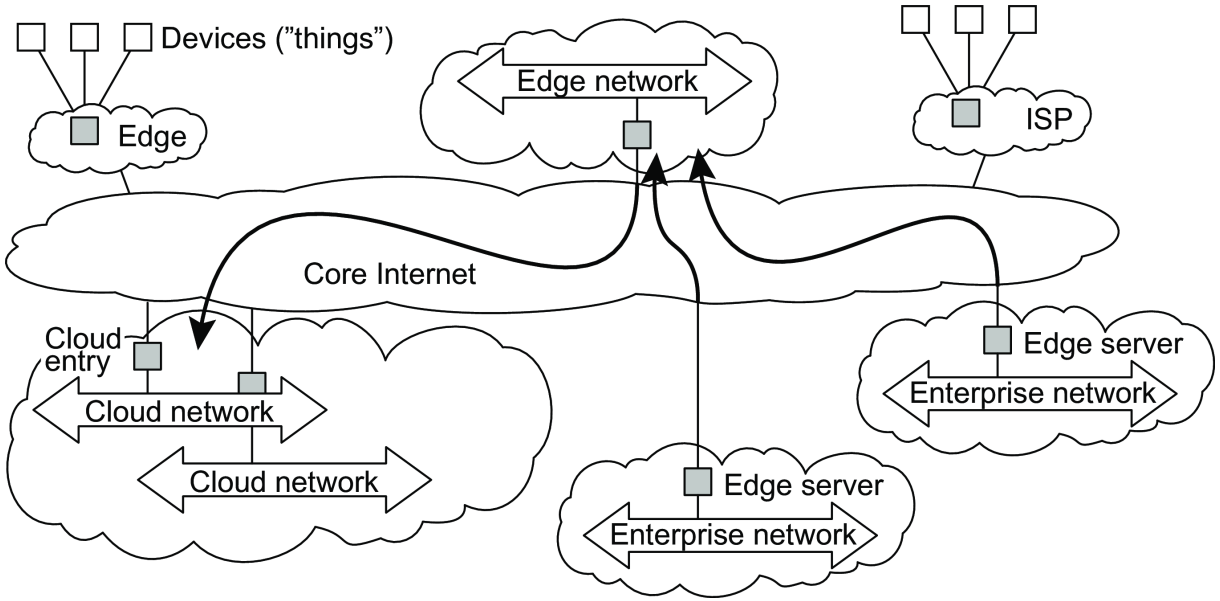


Cloud Computing

- **Hardware:** Processors, routers, power and cooling systems.
  - Customers normally never get to see these.
- **Infrastructure:** Deploys virtualization techniques.
  - Evolves around allocating and managing virtual storage devices and virtual servers.
- **Platform:** Provides higher-level abstractions for storage and such.
  - Example: Amazon S3 storage system offers an API for (locally created) files to be organized and stored in so-called buckets.
- **Application:** Actual applications, such as office suites (text processors, spreadsheet applications, presentation applications).
  - Comparable to the suite of apps shipped with OSes.

Edge-server Architecture

- Systems deployed on the Internet where servers are placed at the edge of the network:
  - the boundary between enterprise networks and the actual Internet.



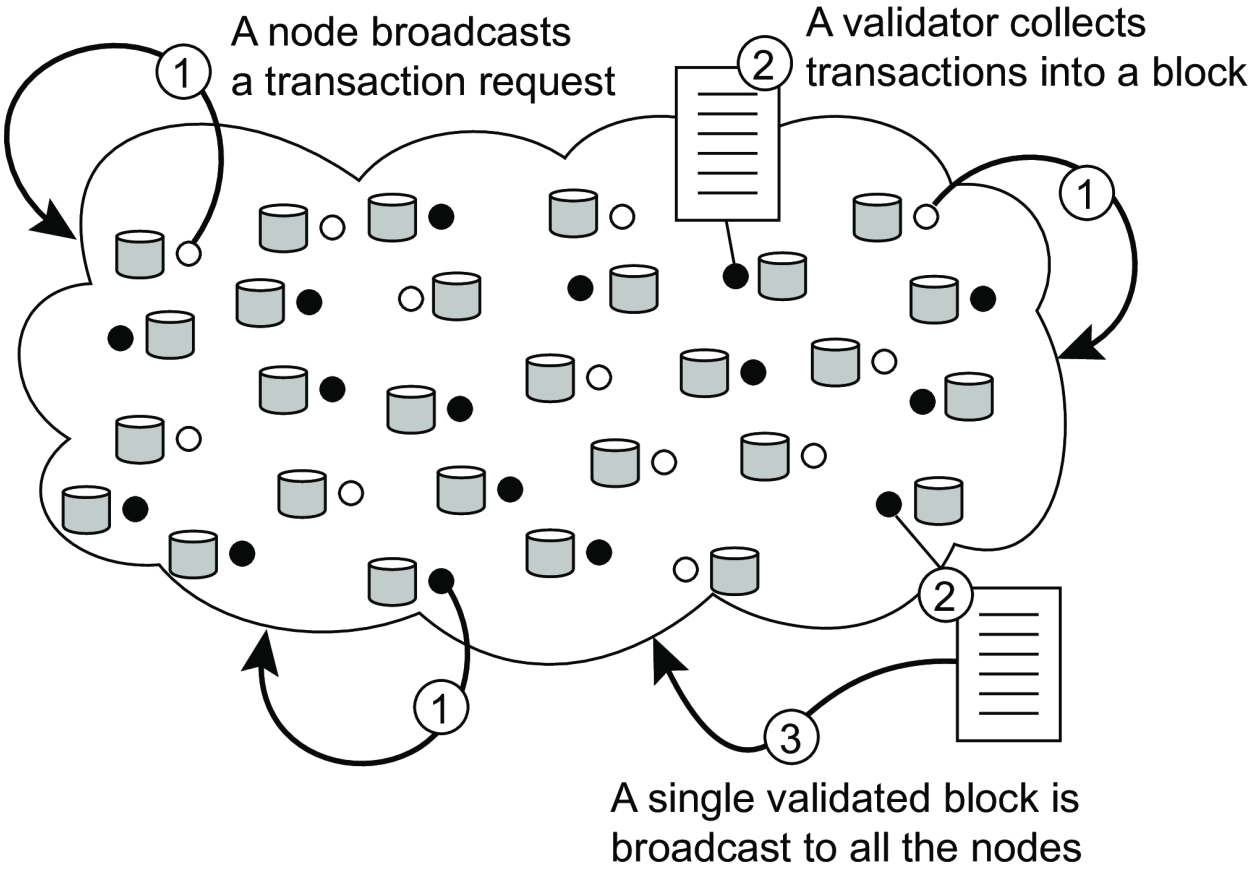
Advantages of Edge Infrastructure

- **Latency and bandwidth:** Especially important for certain real-time applications, such as augmented/virtual reality applications. Many people underestimate the latency and bandwidth to the cloud.
- **Reliability:** The connection to the cloud is often assumed to be unreliable, which is often a false assumption. There may be critical situations in which extremely high connectivity guarantees are needed.
- **Security and privacy:** The implicit assumption is often that when assets are nearby, they can be made better protected. Practice shows that this assumption is generally false. However, securely handling data operations in the cloud may be trickier than within your own organization.

Edge Orchestration

- Managing resources at the edge may be trickier than in the cloud .
  - **Resource allocation:** we need to guarantee the availability of the resources required to perform a service.
  - **Service placement:** we need to decide when and where to place a service. This is notably relevant for mobile applications.
  - **Edge selection:** we need to decide which edge infrastructure should be used when a service needs to be offered. The closest one may not be the best one.

Blockchains

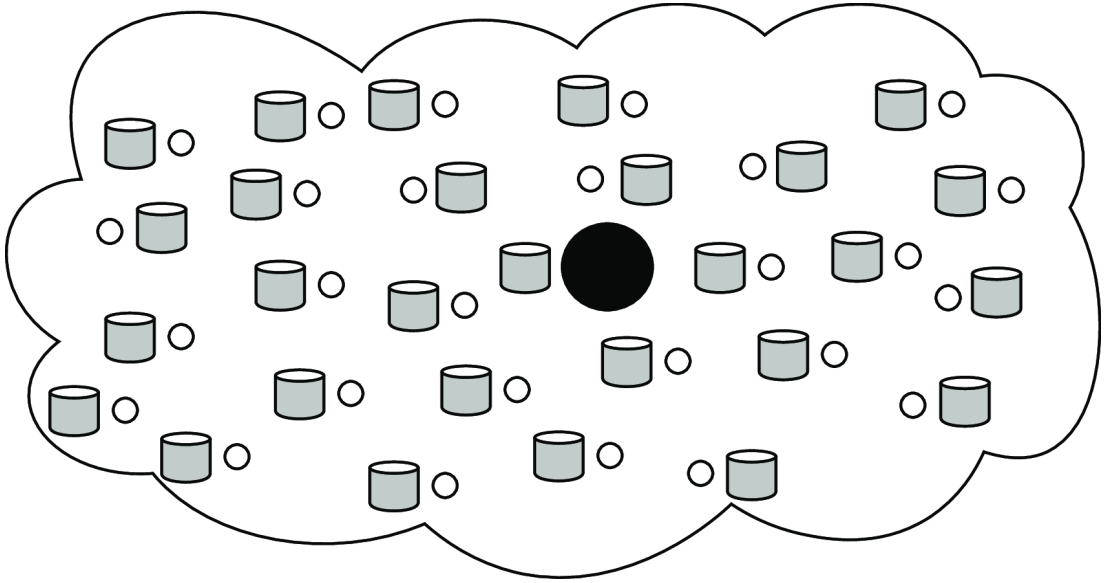


Blockchains

- Principle working of a blockchain system:
  - Blocks are organized into an unforgeable append-only chain
  - Each block in the blockchain is immutable ⇒ massive replication
  - The real snag lies in who is allowed to append a block to a chain!

Appending a Block: Distributed Consensus

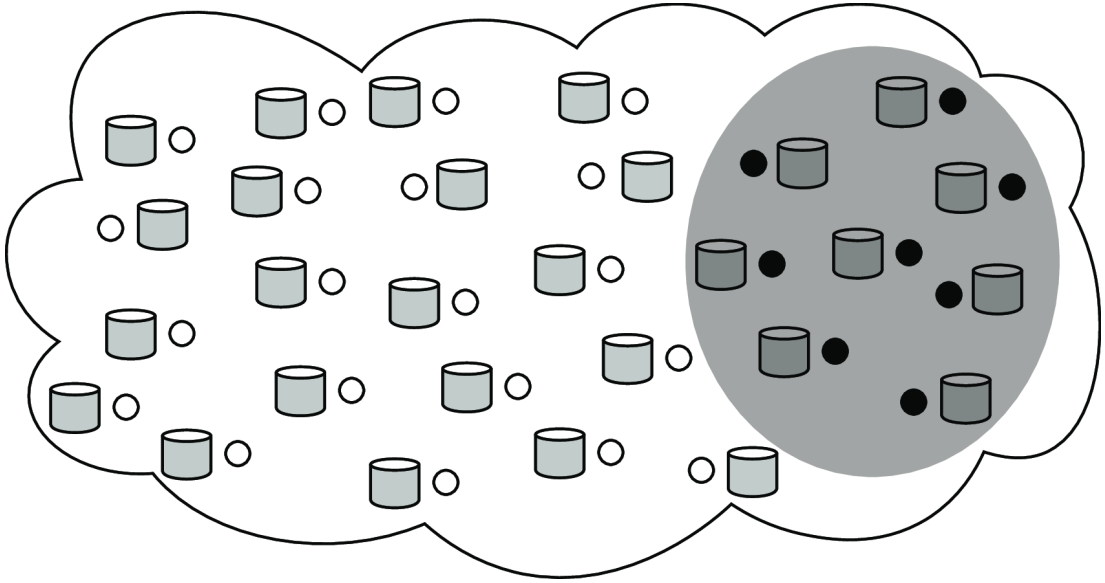
- Centralized solution:



- A single entity decides on which validator can go ahead and append a block.
- Does not fit the design goals of blockchains.

Appending a Block: Distributed Consensus

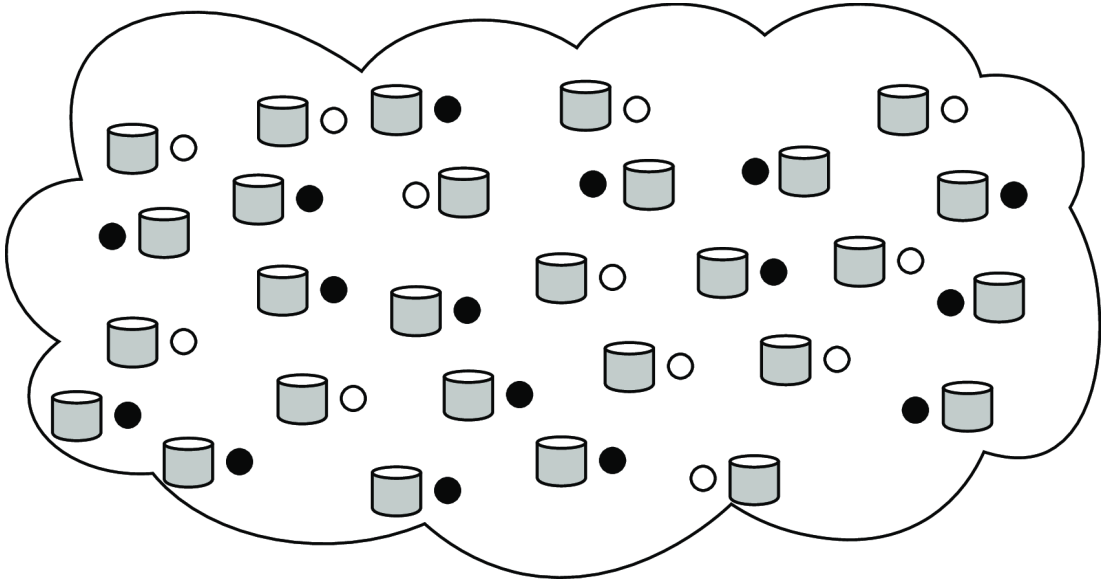
- Distributed solution (permissioned):



- A selected, relatively small group of servers jointly reach consensus on which validator can go ahead.
- None of these servers needs to be trusted, as long as roughly two-thirds behave according to their specifications.
- In practice, only a few tens of servers can be accommodated.

Appending a Block: Distributed Consensus

- Decentralized solution (permissionless):



- Participants collectively engage in a leader election. Only the elected leader is allowed to append a block of validated transactions.
- Large-scale, decentralized leader election that is fair, robust, secure, and so on, is far from trivial.

Chapter Wrap-Up

In the next chapter, we will discuss how the different types of processes play a crucial role in distributed systems. We also take a look at below topics:

- Threads,
- Virtualization,
- Clients,
- Servers,
- Code Migration.